Engineering Team Charter

1. Team Name

Project Week 1 – Transistor Group 12

2. Mission Statement

The primary team purpose is to learn about the fundamentals and use cases of transistors in modern day technologies. We will explore the physics behind bipolar and metal-oxide transistors, why the characteristics of a transistor are important, and how digital systems utilise these characteristics to create complex circuits.

3. Goals & Objectives

- Goal 1: Implementing a functional half adder on a breadboard using transistors by Thursday afternoon, with the option to combine it into a full adder with another team
- Goal 2: Creating and delivering a presentation about transistors to a non-technical audience, with a 3 minute video of the presentation from each member of the group. Finished by Thursday afternoon.
- Goal 3: Utilising proper project management, time management, quality management, change management and sustainability techniques throughout the project. This includes, completing a team charter, Gannt chart, skills matrix. These should be completed by Thursday afternoon.

4. Team Members & Roles

Name	Role	Responsibilities
Ale	Testing Engineer	Quality Management System / breadboard prototyping
Amy	Systems Engineer	Skills Matrix / breadboard prototyping
Luke	Prototype Engineer	Work packages / breadboard prototyping
Dan	Simulation Engineer	Team Charter / breadboard prototyping
Rishabh	Project Manager	Gantt Chart / breadboard prototyping
Mikel		

5. Resources & Constraints

- Tools: Microsoft OneDrive/Office, Project Libre, breadboard, electronic components (resistors, transistors, wires etc), LTspice (simulation)
- Budget: N/A (no money spent)
- Constraints: Design is limited to breadboard

6. Operating Norms

- Daily meetings + design sessions in the lab
- Communication: Teams for messages
- Documentation: All changes logged in quality management system

7. Decision-Making Process

- Technical decisions: Majority vote within the team
- Management decisions: Decided by whoever is working on that particular skill

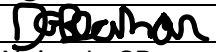
8. Conflict Resolution

- Step 1: Direct discussion between involved parties
- Step 2: Mediation by rest of team
- Step 3: Escalation to speaking to lecturer if unresolved

9. Team Values & Culture

- Transparency
- Continuous improvement
- Inclusivity
- Accountability
- Controversy with civility
- Organisation

10. Sign-Off & Commitment

Name	Signature	Date
Daniel Beaman		02/11/2026
Ale Schierenbeck Robles	AlejandroSR	02/11/2026
Luke Henry		
Mikel Leese		

Rishabh Roy		02/11/2026
Amy Lee		